A Comprehensive Time Series Examination of S&P 500 and NASDAQ Composite Data (2010–2025)

Abstract

This report presents a detailed time series analysis of daily financial data from the S&P 500 (GSPC) and NASDAQ Composite (IXIC) indices spanning 2010–2025. The study includes dataset characterization, detrending procedures, autocorrelation and cross-correlation analysis, and spectral decomposition. The results reveal strong non-stationarity, significant temporal dependencies, synchronous co-movement between indices, and dominant low-frequency components indicative of long-term market trends.

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1. Introduction

Financial time series often exhibit complex behaviors such as trends, volatility clustering, and long-range dependencies. Understanding these characteristics is essential for forecasting, risk modeling, and market structure analysis. This report analyzes two major U.S. equity indices using statistical and spectral methods to uncover underlying temporal patterns.

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2. Dataset Description

The dataset contains **4024 daily observations** from **January 4, 2010 to December 31, 2025**, covering:

- S&P 500 (GSPC): Open, High, Low, Close, Volume, Adjusted
- NASDAQ Composite (IXIC): Open, High, Low, Close, Volume, Adjusted

The report notes: “RangeIndex: 4024 entries, 0 to 4023” and “Data columns (total 13 columns).” All price and volume variables are numeric (float64), while the Date column is converted to a datetime index.

Descriptive statistics show substantial growth in both indices over the 15-year period, with GSPC.Open ranging from ~1027 to ~6936 and IXIC.Adjusted reaching nearly 25,000. These values reflect long-term market expansion and volatility.

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3. Detrending and Stationarity

3.1 Visual Trend Inspection

Plots of GSPC.Adjusted and IXIC.Adjusted reveal:

- Strong upward trends
- Increasing volatility in later years
- No clear seasonal patterns

The report states: “Both … exhibit a clear upward trend … indicating non-stationarity.”

3.2 Detrending Method

To address non-stationarity, **first-order differencing** was applied:

- `GSPC.Adjusted_diff = GSPC.Adjusted.diff()`
- `IXIC.Adjusted_diff = IXIC.Adjusted.diff()`

This transformation removes the deterministic trend and stabilizes the mean. The report confirms: “The strong upward trends … are now largely absent.”

3.3 Remaining Issues

Despite successful detrending, the report notes: “Variance does not appear constant … suggesting heteroscedasticity,” a common feature in financial returns.

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4. Autocorrelation and Cross-Correlation Analysis

4.1 ACF and PACF of Original Series

The ACF for both indices shows:

- Extremely high positive autocorrelation
- Slow decay across 50+ lags

This indicates a **unit root process** and strong trend behavior.

The PACF shows:

- A dominant spike at lag 1
- Near-zero values afterward

This pattern is consistent with an AR(1)-like structure after differencing.

4.2 ACF and PACF After Differencing

Differenced series exhibit:

- Rapidly decaying ACF
- PACF significant only at lag 1 or 2

This confirms that differencing successfully induces stationarity.

4.3 Cross-Correlation Between GSPC and IXIC

The report highlights: “Very high positive cross-correlation centered at lag 0.”

Interpretation:

- The indices move synchronously
- Shared macroeconomic drivers dominate
- High correlations at many lags are due to shared non-stationary trends

After detrending, one would expect a sharper, more localized peak at lag 0.

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5. Spectral Analysis

A periodogram of GSPC.Adjusted reveals:

- A massive spike at extremely low frequencies
- Rapid decline in spectral power at higher frequencies
- No significant periodic components

The report states: “A very high power spectral density at extremely low frequencies … no strong periodic components.”

Interpretation:

- The series is dominated by long-term trend behavior
- No meaningful weekly, monthly, or annual cycles are present
- Spectral results reinforce ACF/PACF findings

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6. Conclusion

This analysis demonstrates that both the S&P 500 and NASDAQ Composite indices exhibit strong non-stationarity, long-term upward trends, and synchronous co-movement. First-order differencing effectively removes the trend, revealing short-memory dynamics suitable for ARMA/ARIMA modeling. Spectral analysis confirms the dominance of low-frequency components and the absence of strong periodic cycles. These findings provide a foundation for further modeling, including volatility analysis and multivariate forecasting.

References

- Extracted content from the uploaded report (TSA_Proposal.pdf)